

OSDP 3.0 Proposed PIV Enhancements

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Changelog

2026-05-30 Added source-backed VCI and PIV Auto examples, clarified PIV Auto interoperability requirements, and improved reviewability:

- Added NIST-derived VCI and PIV APDU fixture coverage, GSA-backed demonstration keys, and reproducible worked examples for PIV Auto and VCI validation. RSA-1024 material is retained only as informative legacy/non-current test-vector material.
- Clarified PIV Auto cryptographic inputs and response handling, including KMAC-derived challenge construction, RSA and ECC GENERAL AUTHENTICATE templates, nonzero UUID requirements, and complete returned 0x7C Dynamic Authentication Templates in `osdp_PIVSTATUSR`.
- Aligned related command semantics for capability advertising, `osdp_CARDSTATUS` errors, selective `osdp_PIVDATA` replies, `osdp_PIVSPE` status/abort modes, and VCI trust-anchor inventory.
- Reworked Appendix C and the references section for standards review: examples are explicitly informative, vector provenance is documented, source fixtures are framed as public test material, and normative references now include the needed APDU, KMAC, X.509, and requirement-language specifications.
- Added supporting scripts and tests for deterministic simulation, fixture replay, VCI import/validation, Appendix C generation, YubiKey material loading, live APDU capture, and reproducible PDF/table builds.

2026-05-27 Version renamed from OSDP 2.3 to OSDP 3.0 throughout. Consolidated PIV Auto, biometric-result, VCI trust-anchor, and SPE updates:

- Removed `osdp_PIVSPE` flag 0x01 (retry continuation); each `osdp_PIVSPE` command now initiates exactly one PIN-entry session and `osdp_PIVSPER` terminates the session unconditionally, including on PIN mismatch. ACUs wishing to allow additional attempts **MAY** re-issue `osdp_PIVSPE`, using the *Attempts Remaining* count from the preceding `osdp_PIVSPER` to guide retry policy. Bit 0x01 is now reserved.
- Aligned `osdp_PIVSPER` and `osdp_PIVBIOMATCHR` result codes for credential removal, changed `osdp_PIVBIOMATCHR` 0x03 to PIN verification required, reserved `osdp_PIVBIOMATCHR` 0x07, clarified biometric assurance limits, and specified that only off-card biometric match success returns the 32-byte SHA-256 hash of tag 0xBC.
- Clarified the Function Code 14 biometric-support reference for `osdp_PIV_LOADTPK` and added CAC as PIV mode data model 0x03.
- Added PIV Auto KMAC challenge derivation, required series and sequence counters, fixed-width supplemental entropy, and the `osdp_PIVSTATUSR` poll reply for reporting credential identity and signed challenge responses. PIV Auto is configured through `osdp_PIVMODE` and is restricted to Secure Channel 2.0 sessions that are not using factory keys.
- Replaced the SP 800-73 Intermediate CVC anchor model with an OSDP-defined VCI trust anchor record (tag 0x7F50) carrying a Profile Identifier (0x80 = 0x01), an eight-byte Issuer

Identifier Number (0x42), and a DER-encoded SubjectPublicKeyInfo (0x7F49). Maximum conforming record size is 576 bytes (0x0240), derived from the RSA-4096 key case.

- Added normative IIN derivation rule: the IIN in `osdp_PIV_VCILOADTA` and in the trust anchor record **SHALL** be the leftmost eight bytes of the subjectKeyIdentifier from the content-signing X.509 certificate.
- Tightened `osdp_PIV_VCILOADTA` multi-part encoding: Anchor Length and Mp-SizeTotal ranges corrected to 0x0000-0x0240; added normative constraint that `MpOffset + MpFragmentSize` **SHALL NOT** exceed Anchor Length; PD **SHALL** verify the reassembled byte string is exactly *Anchor Length* bytes and a single 0x7F50 record before committing.
- Documented variable-length PD storage accounting: PDs **SHALL** account for anchors by their actual *Anchor Length*, not a fixed maximum; replacement is atomic (prior anchor remains if replacement fails); delete releases actual stored bytes.
- Specified the two-tier PD verification path during VCI establishment: direct (trust anchor IIN matches secure messaging CVC IIN, anchor SPKI verifies CVC signature) and intermediate (no direct match triggers retrieval of PIV data object 0x1017/5FC122; Intermediate CVC Subject Identifier must equal CVC IIN, Role 0x5F4C must be 0x12, anchor IIN must match Intermediate CVC IIN, anchor SPKI verifies Intermediate CVC, Intermediate CVC verifies secure messaging CVC). PDs **SHALL NOT** perform X.509 path validation or revocation for VCI trust anchors.
- Added normative ACU post-VCI validation requirements: after `osdp_CARDSTATUSR` reports VCI established, the ACU **SHALL** obtain the content-signing certificate (from cache or via `osdp_PIVDATA` for 0x1017/inner tag 0x70), verify SKI matches *VCI Anchor IIN*, verify SPKI matches the stored anchor record, verify PIV or PIV-I content-signing ECU OID, verify validity period (expired certificate **SHALL** cause session rejection and anchor unload), and perform X.509 path validation and revocation per deployment policy.
- Restored *VCI Trust Anchor ID* to `osdp_CARDSTATUSR` with normative language; added *VCI CVC IIN* and *VCI Anchor IIN* fields (each 8 bytes) so the ACU receives the IIN pair needed to distinguish direct and intermediate verification paths.
- Updated `osdp_PIV_VCITASTATUSR`: clarified that *Available Bytes* reflects variable-length accounting and that the per-record IIN is the anchor selector from the stored trust anchor record, not necessarily the secure messaging CVC IIN reported in `osdp_CARDSTATUSR`.
- Added VCI security preconditions: all VCI-related commands **SHALL** be rejected unless OSDP Secure Channel 2.0 is active and not using factory/default keys; failures reported via `osdp_PDERROR` 0x0006.

2026-01-21 Error-handling cleanup and code assignments: **(1)** `osdp_PIVPUTDATA` now **SHALL** report failures via `osdp_PDERROR` (status word optional) so writes can't silently succeed. **(2)** `osdp_PIVSPE` adds result code 0x09 in `osdp_PIVSPER` for "VCI not established." **(3)** `PDERROR` is credential/global only; PD capability gaps stay `osdp_NAK`. **(4)** Legacy 2.2 replies (e.g., `osdp_CRAUTHR`, `osdp_PIVDATAR`) remain unchanged, with optional follow-up `osdp_PDERROR` detail if the PD has it. **(5)** Documented tentative command/reply codes for the new PIV set:

commands 0xA6 (VCI trust-anchor status) through 0xAF (VCI trust-anchor load), and replies 0x85-0x89 for the new responses.

2026-01-07 Added a PIV Mode interface-mask table aligned to Function Code 13 bits, clarified `osdp_CRAUTHR` reply structure (full GENERAL AUTHENTICATE response without SW1|SW2; status words reported via `osdp_PDERROR` only), tightened `osdp_CARDSTATUS/osdp_CARDSTATUSR` behavior (solicited-only after PD preparation; error on credential removal), remapped the PIV error codes to the 0x10xx block per subcommittee guidance, added PIV APDU references, aligned `osdp_PIVDATA` tag selection to allow Discovery (0x7E)/BIT (0x7F61) subtags in line with SP 800-73-4, clarified that `osdp_PDERROR` conveys a single error per reply, and clarified `osdp_PIV_XMITPAIRING` handling so pairing codes can be provided before VCI is established when the credential requires them.

2025-12-10 Consolidated application-level errors on `osdp_PDERROR` (two-byte codes, optional SW1|SW2 for APDU failures, multi-error support); moved the PIV error set to 0x0100-0x01FF and updated PIV commands to use `osdp_PDERROR` (transport faults remain `osdp_NAK`); made the Global Interface Mask in `osdp_PIVMODE` a required byte (no TLV tag) with an updated field layout; clarified that `osdp_CARDSTATUSR` is solicited-only after PD preparation with legacy poll-based `osdp_RAW` as the aggregated card-present indication; and noted the deprecation of `osdp_GENAUTH` with a recommendation to migrate to `osdp_CRAUTH`.

2025-12-04 Clarified that `osdp_CRAUTHR` returns the full GENERAL AUTHENTICATE data including the Dynamic Authentication Template (excluding only the ISO/IEC 7816 status word).

2025-11-12 Added PIV Biometric Match request/reply and tables (one-bit type values only); moved Enhanced PIV NAKs to a 0x20+ base (added 0x27 Credential Not Present/Removed, 0x28 Biometric Type Not Supported; clarified 0x23 Credential Security Status Not Satisfied); aligned `CRAUTHR` error handling to reference the enhanced NAK set; relocated Enhanced PIV (0x04) and PIV Auto (0x08) indicators to Function Code 12 and cleaned Function Code 13 to interface-only bits (removed non-interface flags; removed PIV Auto from Function Code 14); updated `CARDSTATUS` gating to Function Code 12 bit 0x04; added a PIV Mode Flags table and clarified that any legacy AID lists cited were from prior proposals (not OSDP 2.2); enabled PDF TOC hyperlinks and improved table/caption pagination.

2025-10-30 Clarified `CRAUTH` command/reply behaviour (including RSA, AES, and ECC support), overhauled PIV Mode configuration with TLV-based application and interface controls (including PIV Auto), added card status request/reply messages, expanded PIV data retrieval semantics, introduced PIV Put Data, PIV Secure PIN Entry, TWIC privacy-key load, VCI pairing transmit, and VCI trust-anchor load/status commands, and documented the enhanced error-code set (including VCI/PIV Auto conditions).

2025-09-17 Major PIV Mode and SPE revisions.

2025-06-25 Added multi-AID usage and VCI support notes.

2025-05-27 New function code definitions for 3.0 biometric handling.

1 Introduction

This document captures proposed enhancements to the SIA Open Supervised Device Protocol (OSDP) needed to carry FICAM-compliant Personal Identity Verification (PIV) workflows. The material consolidates draft content, reviewer comments, and interoperability feedback collected during the 2025 working sessions and re-expresses it as normative guidance for command authors and device implementers.

The chapters that follow highlight changes required for existing OSDP 2.2 commands, introduce new transactions where no equivalent existed, and provide supplemental examples that reflect current federal credentialing practices. Wherever the proposal diverges from the baseline specification, the text records both the behavioral requirements and the error handling expectations so that access control units (ACUs) and peripheral devices (PDs) can adopt the profile without relying on vendor extensions.

Legacy compatibility: when a PD encounters a credential-side error on a legacy 2.2 reply (for example `osdp_CRAUTHR` or `osdp_PIVDATAR`), it continues to return the legacy reply shape (often with `DATA_LEN = 0`) and may optionally send an `osdp_PDERROR` with additional detail on a subsequent `osdp_POLL`. ACUs are expected to tolerate or ignore the optional `osdp_PDERROR`; OSDP 2.2 ACUs may log the unknown reply code and proceed.

2 Changes to OSDP 2.2

This section documents the normative deltas that apply when an OSDP 2.2 implementation adopts the Enhanced PIV profile. Each subsection references the original command definition and records the precise behavioral updates, encoding clarifications, and new error handling rules introduced by this proposal.

2.1 Updated Command Definitions

2.1.1 Authentication Challenge (osdp_CRAUTH)

OSDP 2.2 transports the ISO/IEC 7816-4 GENERALAUTHENTICATE command through osdp_CRAUTH. The OSDP 3.0 profile maintains the same framing fields (TOTAL, OFFSET, DATA_LEN) but clarifies how the command payload maps to the PIV operation selected by the algorithm and key reference.

- The *Algorithm* field continues to be sent in the first fragment only. Codes 0x05, 0x06, and 0x07 select RSA private-key operations and preserve backwards compatibility with OSDP 2.2. Codes 0x08, 0x0A, and 0x0C select AES encryption using 128-, 192-, and 256-bit keys respectively. Codes 0x11 and 0x14 identify ECC P-256 and P-384 mechanisms; the selected PIV key reference determines whether the mechanism is used for ECDSA signature generation or ECDH key establishment.
- The *Key* field retains its role as the card's reference identifier. PDs **SHALL NOT** reinterpret or remap the identifier.
- For RSA algorithms (0x05, 0x06, 0x07), the *Challenge* field carries the encoded datum for the RSA private-key operation. The PD forwards the value unmodified; padding and digest encoding are the responsibility of the ACU.
- For AES algorithms (0x08, 0x0A, 0x0C), the *Challenge* field contains the plaintext block(s) to be encrypted. PDs **SHALL** send the data to the credential without padding; ACUs **SHALL** provide data aligned to the block size dictated by the selected key length and card profile.
- For ECC signature operations using key references 0x9A, 0x9C, or 0x9E with algorithms 0x11 or 0x14, the *Challenge* field carries the pre-hashed input to be signed. The PD **SHALL** issue GENERALAUTHENTICATE with command data encoded as 7C Len 82 00 81 Len Challenge; the credential returns the ECDSA signature in tag 0x82. Exponentiation tag 0x85 **SHALL NOT** be used with 0x9A, 0x9C, or 0x9E.
- For ECC key-establishment operations using key reference 0x9D or retired key-management references, algorithms 0x11 and 0x14 use the *Challenge* field to convey the peer public key. The PD **SHALL** issue GENERALAUTHENTICATE with command data encoded as 7C Len 82 00 85 Len (04 || X || Y), where X and Y are the uncompressed coordinates for the selected curve and are left-padded with 0x00 as required. The credential performs elliptic-curve Diffie-Hellman in accordance with SP 800-73.

Table 1: Authentication Challenge (osdp_CRAUTH)

Size	Name	Meaning	Value
1	CMND	Command identifier.	0xA5
2	TOTAL	Length of the complete message, least-significant byte first.	0x0000–0xFFFF
2	OFFSET	Offset of this fragment within the complete message, least-significant byte first.	0x0000–0xFFFF
2	DATA_LEN	Length of the fragment payload, least-significant byte first.	0x0000–0xFFFF
1	Algorithm	Cryptographic mechanism. Supported values include 0x05/0x06/0x07 (RSA private-key operation), 0x08/0x0A/0x0C (AES encryption with 128/192/256-bit keys), and 0x11/0x14 (ECC P-256/P-384; operation depends on the key reference).	0x00–0xFF
1	Reference Identifier	Key reference passed to GENERALAUTHENTICATE.	0x00–0xFF
0-n	Challenge	For RSA algorithms, the encoded datum for the private-key operation. For AES algorithms, the plaintext block(s) to be encrypted, pre-padded by the ACU as required. For ECC 0x9A/0x9C/0x9E, the pre-hashed signature input in tag 0x81; for ECC 0x9D/retired key-management references, the peer public key in tag 0x85.	0x00–0xFF

Deprecated osdp_GENAUTH The legacy osdp_GENAUTH command is deprecated and expected to be removed in a future revision of this specification. Using a PIV credential to decrypt or process arbitrary data is unsafe and unnecessary for credential verification; implementations **SHOULD** use osdp_CRAUTH flows instead.

2.1.2 Response to Challenge (osdp_CRAUTHR)

The reply continues to use the OSDP 2.2 multi-part format. Successful execution yields a response containing the credential's entire GENERALAUTHENTICATE data (minus the ISO/IEC 7816 status word), including the returned Dynamic Authentication Template; PDs **SHALL NOT** strip outer BER-TLV tags or lengths. If the credential reports an error, the PD **SHALL** return the legacy osdp_CRAUTHR reply (for example with DATA_LEN = 0) to preserve OSDP 2.2 behaviour. When available, the PD **SHOULD** also return osdp_PDERROR on a subsequent osdp_POLL with an applicable PIV code—for example 0x1020 (function not supported by credential), 0x1022 (insufficient credential storage), 0x1023 (credential security status not satisfied), 0x1024 (data object/tag not found), 0x1025 (VCI not established), or 0x1027 (credential not present/removed). Generic parameter/security preconditions may continue to use 0x0005 (incorrect parameter) and 0x0006/0x0007 (PIN verification failed/timeout) where those codes remain canonical. ACUs **SHALL** tolerate an optional osdp_PDERROR following the legacy reply; OSDP 2.2 ACUs may log and ignore the unknown reply code.

Reply structure

osdp_CRAUTHR **SHALL** return the credential's full ISO/IEC 7816 GENERALAUTHENTICATE response payload, including the Dynamic Authentication Template, framed by the standard OSDP multi-part fields. PDs **SHALL NOT** include SW1|SW2 in osdp_CRAUTHR; any status words from the credential **SHALL** be carried only in osdp_PDERROR when a card/APDU failure occurs. PDs **SHALL NOT** strip outer BER-TLV tags or lengths from the returned payload.

Table 2: Response to Challenge (osdp_CRAUTHR)

Size	Name	Meaning	Value
1	CMND	Reply identifier.	0x82
2	TOTAL	Length of the complete response, least-significant byte first.	0x0000–0xFFFF
2	OFFSET	Offset of this fragment within the complete response, least-significant byte first.	0x0000–0xFFFF
2	DATA_LEN	Length of the fragment payload, least-significant byte first.	0x0000–0xFFFF
0-n	Response Data	Signature, encrypted block, or shared-secret material returned by the credential. The credential's BER-TLV structure is preserved exactly as received (status word excluded).	0x00–0xFF

2.1.3 Get PIV Data (osdp_PIVDATA)

OSDP 2.2 defined `osdp_PIVDATA` with a five-byte payload that selected a PIV object and optional element. The original text did not explicitly specify how to pad identifiers shorter than three bytes, how nested tags should be returned, or how to interpret the data offset. OSDP 3.0 clarifies these behaviors so that ACUs and PDs apply a single normative approach.

- The ACU **SHALL** transmit the PIV Object Identifier as three bytes encoded most-significant byte first. When the identifier contains fewer than three significant bytes, the ACU **SHALL** left-pad the value with 0x00 or 0x0000, as required, so that the transmitted field is exactly three bytes. The PD **SHALL NOT** forward those pad bytes to the credential; it **SHALL** issue the ISO 7816-4 GET DATA APDU using only the significant identifier bytes. If VCI secure messaging is established, the PD **SHALL** use the appropriate secure channel automatically; no explicit selector is present in the command.
- The field previously described as “PIV element ID” is renamed *PIV Tag*. A tag value of 0x00 directs the PD to return the complete BER-TLV structure, including the native outer tag (for example 0x7E Discovery, 0x7F61 BIT Group Template) or the 0x53/0x73 wrapper used by other PIV objects. Any non-zero value selects a single root-level tag within the object. When the outer tag is 0x53 or 0x73, the PD **SHALL** strip that wrapper before returning the selected tag. For other outer tags, the PD **SHALL** return the requested tag if present and **SHALL NOT** reject the request solely because the wrapper is not 0x53/0x73. Tags **SHALL** be a single byte in the range 0x01-0xFF; multi-byte DER tags are not supported and **MUST** be retrieved by issuing a tag value of 0x00 and parsing the returned object at the ACU.
- The data offset now supports one- or two-byte encodings. PDs **SHALL** accept either form and determine which was used based on the command length. The offset applies to the selected payload (entire object when the tag is zero, or the chosen tag otherwise). When a credential-side error occurs, PDs **SHALL** return the legacy `osdp_PIVDATAR` reply (for example with `DATA_LEN = 0`) to maintain OSDP 2.2 compatibility. If available, PDs **MAY** report additional detail via `osdp_PDERROR` on a subsequent `osdp_POLL` using code 0x1023 when security prerequisites are not satisfied, 0x1024 when the requested object or tag is absent, or 0x1027 when the credential is not present or is removed; see Subsection 2.2. ACUs **SHALL** tolerate an optional `osdp_PDERROR` following the legacy reply; OSDP 2.2 ACUs may log and ignore the unknown reply code.

Table 3: Get PIV Data (osdp_PIVDATA)

Size	Name	Meaning	Value
1	CMND	Command identifier.	0xA3
3	PIV Object Identifier	Three-byte identifier of the PIV data object per SP 800-73-4, encoded MSB first. Leading zero bytes SHALL be supplied by the ACU when the identifier is shorter than three bytes; the PD SHALL drop those pad bytes before issuing the APDU.	0x00–0xFF ¹
1	PIV Tag	Single-byte ASN.1 DER tag of a root-level element. Tag value 0x00 selects the entire object; non-zero tags select a root-level element and require the PD to strip the outer 0x53/0x73 wrapper, verify the length, and return only the tagged value. Multi-byte tags are not directly addressable and SHALL be retrieved by requesting tag 0x00.	0x00; or 0x01–0xFF
1-2	Data Offset	Byte offset within the requested content. Single-byte and two-byte encodings SHALL be accepted; the PD determines the format from the packet length.	0x00–0xFFFF

¹ Each position MAY carry any value from 0x00 to 0xFF; the ACU pads on the left with 0x00 when the identifier has fewer than three significant bytes.

2.1.4 PIV Data Reply (osdp_PIVDATAR)

The reply structure is unchanged from a field-count perspective, but OSDP 3.0 aligns the semantics with the command refinements above:

- TOTAL, OFFSET, and DATA_LEN continue to be encoded LSB first; they now explicitly report the size, fragment base, and fragment length for the content selected by the osdp_PIVDATA request (entire object or chosen tag).
- CARD_DATA **SHALL** convey the exact bytes obtained from the credential after applying any tag stripping specified by the command. No additional framing or padding is permitted.

Table 4: PIV Data Reply (osdp_PIVDATAR)

Size	Name	Meaning	Value
1	CMND	Reply identifier.	0x80
2	Total Length	Length in bytes of the complete data object or tagged element, least-significant byte first.	0x0000–0xFFFF
2	Fragment Offset	Offset in bytes of this fragment within the complete payload, least-significant byte first.	0x0000–0xFFFF
2	Fragment Length	Length in bytes of the data carried in this fragment, least-significant byte first.	0x0000–0xFFFF
0-n	Card Data	Requested data bytes exactly as returned by the credential. When a tagged element was selected, this sequence contains the complete selected BER-TLV, including tag, length, and value.	0x00–0xFF

2.2 PD Error Codes for PIV

PD error codes are carried in osdp_PDERROR using two-byte codes (least-significant byte first). The common PD codes remain available (for example 0x0001 SC_ERROR with optional SW1|SW2, 0x0002 SC_TIMEOUT, 0x0004 TLV_MISSING, 0x0005 INVALID_PARAM, 0x0006 SECURE_CHANNEL_REQUIRED, 0x0007 MP_INVALID, 0x0008 MP_OUT_OF_SEQUENCE, 0x0009 MP_OUT_OF_BOUNDS). The following PIV-specific codes occupy the 0x1000-0x10FF range:

Table 5: Enhanced PIV Error Codes

Code	Description
0x1020	Function not supported by credential (PD capability gaps SHALL use osdp_NAK function-not-supported).
0x1021	Insufficient PD memory or storage to complete the requested operation.
0x1022	Insufficient credential memory or storage space.
0x1023	Credential security status not satisfied.
0x1024	Data object or tag not found.
0x1025	VCI not established.
0x1027	Credential not present or removed.

2.2.1 PD Application Error Reply (osdp_PDERROR)

PDs **SHALL** use osdp_PDERROR only for credential-side or other global application errors that are not already defined in a command-specific reply. The error code is two bytes, encoded least-significant byte first. When a card/APDU failure occurs (for example SC_ERROR or a PIV APDU failure), the PD **MAY** append the ISO/IEC 7816 status word (SW1|SW2) when it is available to the application layer; otherwise the status word is omitted. A single error is reported per osdp_PDERROR reply. Transport-level faults (framing, CRC, sequence) continue to use osdp_NAK.

Table 6: PD Application Error Reply (osdp_PDERROR)

Size	Name	Meaning	Value
1	CMND	Command identifier.	0x84
2	Error Code	Two-byte error code, least-significant byte first.	0x0001–0x10FF
0 or 2	Status Word	Optional ISO/IEC 7816 status word for card/APDU failures; present only when the PD can surface it to the application layer.	–

2.3 Updated Function Codes

2.3.1 Function Code 12 – Smart Card Support

Function Code 12 advertises smart-card transport capabilities and profile support at the PD. The compliance byte uses the following bit assignments:

Table 7: Function Code 12 Compliance Flags

Mask	Description
0x01	PD supports transparent reader mode.
0x02	PD supports extended packet mode.
0x04	PD supports the OSDP 3.0 Enhanced PIV profile, including osdp_PIVDATA refinements and related commands.
0x08	PD supports OSDP 3.0 autonomous PIV operation (PIV Auto).

Bits not listed are reserved and **SHALL** be transmitted as zero.

2.3.2 Function Code 13 – Reader Interfaces

OSDP 2.2 defined Function Code 13 as a reader count indicator with a compliance byte fixed at 0x00. Enhanced PIV-capable PDs **SHALL** continue to populate the reader count field, but they **MAY** set bits within

the compliance byte to advertise the downstream credential technologies they expose. A value of 0x00 remains valid and retains the legacy meaning of “interface presence unspecified.”

When a PD reports specific interfaces, it **SHALL** encode the compliance byte using the flags in Table 8. The reader count byte continues to report the number of attached credential interfaces.

Table 8: Function Code 13 Compliance Flags

Mask	Description
0x01	PD presents a contact card interface (ISO 7816 or equivalent).
0x02	PD presents a high-frequency contactless credential interface (13.56 MHz NFC).
0x04	PD presents a low-frequency credential interface (approximately 125 kHz prox).
0x08	PD supports Virtual Contact Interface (VCI) secure messaging for applicable modes.
0x10	PD presents a Bluetooth credential interface.
0x20	PD presents a barcode credential interface.
0x40	PD presents an ultra-high-frequency credential interface (UHF/RAIN RFID).

All other bits are reserved for future use and **SHALL** be transmitted as zero.

2.3.3 Function Code 14 – Biometric Support

Function Code 14 remains backward compatible with OSDP 2.2. A compliance value of 0x00, 0x01, 0x02, or 0x03 retains its original meaning and indicates that the PD does not support the OSDP 3.0 biometric command set. Devices that support the enhanced commands **SHALL** set Bit 7 (0x80). When Bit 7 is asserted, the remaining bits of the compliance byte are interpreted as a capability bitfield. Bits not explicitly defined in Table 9 are reserved and **SHALL** be transmitted as zero.

Table 9: Function Code 14 Compliance Flags

Mask	Description
0x80	PD supports the OSDP 3.0 biometric command set.
0x01	PD performs off-card fingerprint comparison using FIPS 201 BIO ANSI INCITS 378 templates.
0x02	PD performs on-card fingerprint comparison using FIPS 201 OCC processes.
0x04	PD performs off-card iris comparison aligned with FIPS 201 BIO guidance.
0x08	PD performs off-card facial comparison using ANSI INCITS 385-2004 images stored on the credential.

3 New or Enhanced PIV Commands

This section introduces commands defined exclusively for the OSDP 3.0 Enhanced PIV profile.

3.1 PIV Put Data (osdp_PIVPUTDATA)

The `osdp_PIVPUTDATA` command (command code 0xA8) instructs the PD to deliver an ISO 7816-4 PUT DATA operation to the credential. The ACU **SHALL** construct the payload as a complete BER-TLV object, including the desired outer tag (0x7E for the Discovery Object, 0x7F61 for a BIT Group Template, 0x5C/0x53 sequences for other data objects, and so on). The PD **SHALL** forward the payload to the credential using CLA = 0x00, INS = 0xDB, P1 = 0x3F, and P2 = 0xFF. If VCI secure messaging is established, the PD **SHALL** select the secure channel automatically and adjust the CLA accordingly. Chained APDUs **SHALL** be generated by the PD when the payload exceeds a single-card APDU.

Table 10: PIV Put Data (osdp_PIVPUTDATA)

Size	Name	Meaning	Value
1	CMND	Command identifier.	0xA8 (tentative)
2	MpSizeTotal	Total size of the complete PUT DATA payload, least-significant byte first.	0x0000–0xFFFF
2	MpOffset	Offset of this fragment within the complete payload, least-significant byte first.	0x0000–0xFFFF
2	MpFragmentSize	Number of payload bytes carried in this fragment, least-significant byte first.	0x0000–0xFFFF
0-n	Data	BER-TLV payload to be written to the credential. This field already includes the outer object tag selected by the ACU.	0x00–0xFF

Responses A successful write is acknowledged with `osdp_ACK`. If the PD does not implement the command, it **SHALL** return `osdp_NAK` function-not-supported. When a credential-side failure occurs, the PD **SHALL** report the failure—either immediately or on a subsequent `osdp_POLL`—using `osdp_PDERROR` and one of the PIV error codes defined in Subsection 2.2: 0x1020 when the function is not supported by the credential, 0x1021 when the PD lacks buffer space to stage the payload, 0x1022 when the credential has insufficient storage, 0x1023 when security conditions are not met, or 0x1027 when the credential is not present or is removed during the operation. The status word is optional.

3.2 Card Status (osdp_CARDSTATUS)

The osdp_CARDSTATUS command (command code 0xA9, tentative) requests the current credential context from the PD. PDs **SHALL** advertise support via Function Code 12 bit 0x04 (Enhanced PIV) before controllers attempt to use this command. The payload selects the status record that should be returned (currently only the standard profile).

Table 11: Card Status Request (osdp_CARDSTATUS)

Size	Name	Meaning	Value
1	CMND	Command identifier.	0xA9 (tentative)
1	Status Type	Requested status record. 0x01 selects the standard PIV card status profile. Values 0x02-0x7F are reserved and shall elicit osdp_PDERROR. Values 0x80-0xFF are reserved for private use.	0x00-0xFF

PDs **SHALL** return osdp_NAK function-not-supported when the command itself is not implemented. If the requested status type is not supported or the payload is malformed, PDs **SHALL** report the condition via osdp_PDERROR (for example 0x0005 incorrect parameter) on a subsequent osdp_POLL.

3.3 Card Status Reply (osdp_CARDSTATUSR)

In response to osdp_CARDSTATUS, the PD reports a ready-to-proceed credential state after completing its card-present workflow (power, application selection, and required reads). PDs **SHALL NOT** send osdp_CARDSTATUSR unsolicited on osdp_POLL. Controllers **SHOULD** continue to use legacy osdp_RAW poll replies as the single aggregated “card presented and data acquired” indication; PIN usage policy and VCI connectivity **SHALL NOT** be included in unsolicited poll indications. Detailed readiness (credential type, selected AID, cached data, and readiness flags) **SHALL** be returned only in solicited osdp_CARDSTATUSR. When VCI is established, the reply reports both the secure messaging CVC IIN and the loaded anchor IIN used by the PD. These values are identical for direct CVC verification and differ when an Intermediate CVC is used. The PD **SHALL** set the *VCI Trust Anchor ID* to the identifier assigned in osdp_PIV_VCILOADTA; this value **SHALL** correspond to an entry in the osdp_PIV_VCITASTATUSR inventory whose IIN matches the *VCI Anchor IIN* field, so that the ACU can correlate reported VCI state with the anchor management record.

Table 12: Card Status Reply (osdp_CARDSTATUSR)

Size	Name	Meaning	Value
1	CMND	Reply identifier.	0x85 (tentative)
1	Detected Credential Type	Current card interface: 0x00 none, 0x01 ISO 7816 contact, 0x02 ISO 14443 contactless, 0x03 other ISO 7816 credential (for example DESFire, Seos), 0x04 low-frequency credential, 0x05 unknown type, 0x06-0x7F reserved, 0x80-0xFF private use.	0x00–0xFF
1	VCI Status	0x00 VCI not established, 0x01 VCI established. Values 0x02-0x7F reserved, 0x80-0xFF private use.	0x00–0xFF
1	VCI Trust Anchor ID	The Trust Anchor ID used by the PD during VCI establishment. The PD SHALL set this field to the identifier assigned in osdp_PIV_VCILOADTA; this value SHALL correspond to an entry in the osdp_PIV_VCITASTATUSR inventory whose IIN matches the <i>VCI Anchor IIN</i> field. Set to 0x00 when VCI is not established.	0x00–0xFF
8	VCI CVC IIN	Issuer Identification Number from the secure messaging CVC tag 0x42. Set to all zero when VCI is not established.	0x00–0xFF
8	VCI Anchor IIN	Issuer Identification Number of the loaded VCI trust anchor used by the PD. Set to all zero when VCI is not established or no anchor was used.	0x00–0xFF
2	PIN Usage Policy	PIN Usage Policy bytes (tag 0x5F2F) returned verbatim when available. A value of 0xFFFF indicates the policy is not present or has not yet been read.	0x0000–0xFFFF
1	Selected AID Length	Length (0-16) of the selected AID. 0x00 indicates no application is currently selected.	0x00–0x10

Size	Name	Meaning	Value
0-16	Selected AID	Application Identifier read from the credential (for example from the FCI template). Present only when the length is non-zero.	0x00–0xFF

A PD that cannot provide the requested status **SHALL** return `osdp_NAK` when the command is unsupported. For credential-side failures encountered while preparing status, the PD **MAY** follow the legacy flow (no `osdp_CARDSTATUSR` data) and, if available, report details via `osdp_PDERROR` on a subsequent `osdp_POLL`. Successful replies include the entire selected AID (when available) and the fixed two-byte PIN Usage Policy field after the PD has completed its preparation. When PIN Usage Policy bytes from the credential's Discovery Object (tag 0x5F2F) are available, the PD **SHALL** report those two bytes verbatim. When no PIV PIN Usage Policy is present or the policy has not yet been read, the PD **SHALL** report 0xFFFF. The VCI fields describe stable readiness (for example whether a secure channel is established) and the IIN evidence the ACU needs for validation; transient negotiation state **SHALL NOT** be reported. If the credential is removed before preparation completes, the PD **MAY** report the condition via `osdp_PDERROR` (for example 0x1027) instead of `osdp_CARDSTATUSR`. `osdp_CARDSTATUSR` is sent only in response to a matching `osdp_CARDSTATUS` request.

3.4 PIV Secure PIN Entry (`osdp_PIVSPE`)

The `osdp_PIVSPE` command (command code 0xAA, tentative) instructs the PD to perform an ISO/IEC 7816 VERIFY operation on the credential, allowing the cardholder to enter a PIN directly on the reader. The ACU is responsible for conveying a clear user indication that secure PIN entry is in progress (for example, via displays or annunciators). While `osdp_PIVSPE` is active, the PD **SHALL** suppress all `osdp_KEYPAD` reports to the ACU and terminate SPE mode only after returning a result via `osdp_PIVSPER`.

Table 13: PIV Secure PIN Entry (`osdp_PIVSPE`)

Size	Name	Meaning	Value
1	CMND	Command identifier.	0xAA (tentative)
1	SPE Mode	Secure PIN Entry operating mode. See Table 14.	0x00–0xFF
1	Start Timeout	Seconds allowed to begin key entry (0 = no timeout).	0x00–0xFF
1	Inter-key Timeout	Seconds allowed between keystrokes (0 = no timeout).	0x00–0xFF
1	Flags	Behaviour modifiers. See Table 15.	0x00–0xFF
2	Cache Validity	PIN cache validity in seconds (LSB first). A value of 0x0000 disables caching.	0x0000–0xFFFF

Table 14: SPE Mode Values

Value	Description
0x00	Status only – report current SPE state without initiating entry.
0x01	Normal – single PIN verification with the credential present (contact).
0x02	VCI pre-cache – collect a PIN prior to credential presentation for VCI use.
0x03	VCI execute – perform PIN verification over VCI using a cached PIN.
0x04	PIN caching – cache PIN for subsequent operations (non-FICAM use cases).
0x05	Clear/abort – clear any cached PIN and abort the current operation.
0x06–0x7F	Reserved for future use (set to zero).
0x80–0xFF	Private use.

Table 15: SPE Flag Values

Mask	Description
0x02	Require an explicit cache-clear command; do not auto-clear cached PINs on credential removal.
0x04	Skip pairing-code verification when the credential and policy permit VCI without a pairing code.
0x08	Auto-execute VCI verification when the credential is presented after a successful pre-cache (Mode 0x02).
0x01, 0x10–0x80	Reserved and SHALL be transmitted as zero.

Controllers select an *SPE Mode* to match the desired workflow (status query, contact PIN verification, VCI pre-cache/execute, PIN caching, or abort). Modes that rely on VCI (for example 0x02 and 0x03) require the PD to have established VCI beforehand. *Start Timeout* and *Inter-key Timeout* values of zero disable the respective timer. *Cache Validity* is expressed in seconds (LSB first); a value of zero disables PIN caching. If a mode requests PIN caching, the PD **SHALL NOT** transmit the PIN to the ACU.

Each `osdp_PIVSPE` command that selects a PIN-entry mode initiates exactly one PIN-entry session. Status-only and clear/abort modes do not initiate PIN entry; they report or clear PD state and then terminate. When the PD returns `osdp_PIVSPER` for a PIN-entry mode, the SPE session terminates unconditionally, including on PIN mismatch, regardless of whether the credential remains present. The ACU **MAY** re-issue `osdp_PIVSPE` to permit an additional attempt; it **SHOULD** use the *Attempts Remaining* count from the preceding `osdp_PIVSPER` to inform its retry policy.

A successful request is acknowledged with `osdp_ACK`. If the PD does not implement the command, it **SHALL** return `osdp_NAK` function-not-supported. Credential-side outcomes (for example credential removal, incorrect parameters, security status not satisfied, or VCI not established) **SHALL** be reported in

the subsequent `osdp_PIVSPER` result code per Table 17. After issuing `osdp_PIVSPE`, the ACU **SHALL NOT** send other PIV commands until a corresponding `osdp_PIVSPER` reply is received.

3.5 PIV Secure PIN Entry Reply (`osdp_PIVSPER`)

The `osdp_PIVSPER` reply (reply code 0x86, tentative) reports the outcome of the most recent `osdp_PIVSPE` command. The PD sends `osdp_PIVSPER` as a response to `osdp_POLL` once SPE completes.

Table 16: PIV Secure PIN Entry Reply (`osdp_PIVSPER`)

Size	Name	Meaning	Value
1	CMND	Reply identifier.	0x86 (tentative)
1	Result	Outcome of the secure PIN entry operation. See Table 17.	0x00–0xFF
1	Attempts Remaining	Match attempts remaining when the result is PIN mismatch; otherwise transmitted as 0x00.	0x00–0xFF

Result codes follow Table 17. When the result indicates a PIN mismatch, the *Attempts Remaining* byte conveys the number of retries still permitted; otherwise the field is reserved and **SHALL** be transmitted as zero.

Table 17: PIVSPE Result Codes

Value	Description
0x00	Success – PIN verified.
0x01	Timeout – PIN entry did not begin within the allotted time.
0x02	Timeout – inter-digit interval exceeded.
0x03	PIN mismatch.
0x04	Credential removed during operation.
0x05	Credential PIN blocked.
0x06	Wrong interface for the requested mode.
0x07	Incorrect parameters.
0x08	Security condition not met.
0x09	VCI not established for the requested mode.
0x0A–0xFF	Reserved for future use.

3.6 PIV Biometric Match (osdp_PIVBIOMATCH)

The `osdp_PIVBIOMATCH` command (command code 0xAB, tentative) instructs the PD to acquire a live sample and compare it to the specified biometric template from the active application. Depending on the *Match Mode*, the comparison occurs either on the credential (on-card match) or within the PD (off-card match).

Table 18: PIV Biometric Match (osdp_PIVBIOMATCH)

Size	Name	Meaning	Value
1	CMND	Command identifier.	0xAB (tentative)
1	Biometric Type	Selects the template type to compare. See Table 19.	0x00–0xFF
1	Match Mode	0x00 off-card match; 0x01 on-card match.	0x00–0x01

Table 19: Biometric Type Values

Value	Description
0x01	Cardholder fingerprint template.
0x02	Cardholder facial image.
0x04	Cardholder iris template.

Controllers set *Match Mode* to select off-card or on-card comparison. Off-card flows require the PD to retrieve the appropriate template and, where applicable (e.g., TWIC contactless), to have a valid privacy key loaded via `osdp_PIV_LOADTPK`. On-card flows require the credential to support OCC. If the PD does not implement the command, it **SHALL** return `osdp_NAK` function-not-supported. Credential-side match outcomes (for example PIN verification required or credential removal during operation) **SHALL** be reported in the subsequent `osdp_PIVBIOMATCHR` result code per Table 21. Conditions that prevent the command from being attempted (for example unsupported credential function, missing VCI when required, or invalid command parameters) **MAY** be reported via `osdp_PDERROR` on a subsequent `osdp_POLL`.

3.7 PIV Biometric Match Reply (osdp_PIVBIOMATCHR)

After `osdp_PIVBIOMATCH` completes, the PD reports the result as a poll response using reply code 0x87 (tentative).

Table 20: PIV Biometric Match Reply (osdp_PIVBIOMATCHR)

Size	Name	Meaning	Value
1	CMND	Reply identifier.	0x87 (tentative)
1	Result	Outcome of the biometric match. See Table 21.	0x00–0xFF
0 or 32	Hash	32-byte SHA-256 hash of the biometric template tag 0xBC data field. The PD SHALL include this field only when <i>Result</i> is 0x00 (off-card match success). The PD SHALL omit this field for on-card match success (0x09) and all failure result codes.	0x00–0xFF

When *Result* is 0x00 (off-card match success), the PD **SHALL** include the 32-byte SHA-256 hash of the biometric template tag 0xBC data field in the reply. For on-card match success (*Result* 0x09) and for all failure result codes, the PD **SHALL** omit the hash field; ACUs **SHALL NOT** rely on a hash received alongside any result other than 0x00.

Off-card biometric comparison demonstrates only that the live sample matched template data presented to the PD. Because off-card biometric objects can be copied or duplicated, an off-card biometric match does not by itself prove authenticity of the credential. On-card biometric comparison is a credential-controlled assertion; the biometric object may be attacker controlled and the PD does not receive the template data being matched. To obtain cryptographic proof associated with an on-card biometric success, the ACU **SHALL** perform an additional authentication using an appropriately verified credential key after the successful on-card comparison.

Table 21: PIVBIOMATCH Result Codes

Value	Description
0x00	Biometric match success (off-card match).
0x01	Biometric match timeout (no entry started).
0x02	Biometric match failed (no match).
0x03	PIN verification required.
0x04	Credential removed during operation.
0x05	PIN verification failed.
0x06	PIN verification timeout.
0x07	Reserved for future use.
0x08	Reserved (PD/credential errors are reported via osdp_PDERROR).
0x09	On-card match success.

Value	Description
0x0A–0x7F	Reserved for future use.
0x80–0xFF	Private use.

3.8 TWIC Privacy Key Load (osdp_PIV_LOADTPK)

The `osdp_PIV_LOADTPK` command (command code 0xAC, tentative) instructs the PD to cache a TWIC Privacy Key (TPK) for use with subsequent biometric operations against a TWIC credential. Function Code 14 is the biometric-support compliance byte; before using this command, controllers **SHOULD** verify that Function Code 14 advertises the OSDP 3.0 biometric command set and the relevant biometric modality bits for the intended TWIC operation.

Table 22: TWIC Privacy Key Load (osdp_PIV_LOADTPK)

Size	Name	Meaning	Value
1	CMND	Command identifier.	0xAC (tentative)
1	Cache Timeout	Number of seconds to retain the cached TPK. 0x00 clears any cached key immediately; 0x01–0xFF permit caching for up to 255 seconds.	0x00–0xFF
1	Control Flags	Bit 0 set requests automatic clearing when the credential is removed; all other bits SHALL be zero.	0x00–0xFF
32	TPK	Trusted Processing Key bytes to cache for the next TWIC biometric operation.	0x00–0xFF

Controllers set *Cache Timeout* to define how long (in seconds) the PD may retain the key. A value of 0x00 clears any cached TPK immediately. Values 0x01–0xFF allow caching for up to 255 seconds. Bit 0 of *Control Flags* directs the PD to clear the cached TPK when the credential is removed; all other bits are reserved and **SHALL** be transmitted as zero. If the PD does not support TWIC TPK caching it **SHALL** return `osdp_NAK` function-not-supported. Invalid payloads (for example reserved bits set) **SHALL** elicit `osdp_PDERROR` 0x0005.

3.9 PIV VCI Pairing Code Transmit (osdp_PIV_XMITPAIRING)

The `osdp_PIV_XMITPAIRING` command (command code 0xAD, tentative) delivers an eight-digit pairing code to the PD for use during Virtual Contact Interface (VCI) establishment. Controllers **SHALL NOT** issue this command unless a PIV application is currently selected and VCI mode is active.

Table 23: PIV VCI Pairing Code Transmit (osdp_PIV_XMITPAIRING)

Size	Name	Meaning	Value
1	CMND	Command identifier.	0xAD (tentative)
8	Pairing Code	Eight ASCII digits ('0'-'9') representing the VCI pairing code associated with the currently selected PIV application.	0x30–0x39

The pairing code is encoded as eight ASCII digits ('0'-'9'). If no PIV application is selected, the PD **SHALL** return `osdp_PDError 0x0005`. Otherwise, the PD **SHALL** accept the pairing code whether or not VCI (Virtual Contact Interface) secure messaging is already established and use it when required by the credential's Discovery Object PIN Usage Policy (for example when bit 4 is set and bit 3 is clear in the first byte). When VCI secure messaging is already active or the credential does not require pairing to establish VCI, the PD **MAY** ignore the supplied code after acknowledging receipt. PDs **SHALL** clear any stored pairing code when VCI is torn down or when the active PIV mode changes.

VCI security preconditions VCI trust-anchor load/status, VCI pairing-code transmit, VCI establishment, and PIV operations that depend on VCI **SHALL** be accepted only when OSDP Secure Channel 2.0 is active between the ACU and PD. A PD **SHALL** reject VCI-related functionality when operating with factory/default SCBK or other factory keys. Security-precondition failures **SHALL** be reported using `osdp_PDError 0x0006` where the command supports `osdp_PDError`; otherwise the command's existing failure reporting path applies.

3.10 Set PIV Mode (osdp_PIVMODE)

The `osdp_PIVMODE` command (command code 0xAE, tentative) defines the operating context for PIV, TWIC, and related credential transactions. OSDP 3.0 replaces the legacy "Application + AID" fields with a configuration payload that includes a required Global Interface Mask byte, PIV Auto challenge parameters when autonomous operation is enabled, and a TLV list of configuration descriptors so controllers can specify multiple application profiles, preferred interfaces, and optional Application Identifiers. PDs **SHALL** advertise support via Function Code 12 bit 0x04 (Enhanced PIV).

Table 24: Set PIV Mode (osdp_PIVMODE)

Size	Name	Meaning	Value
1	CMND	Command identifier.	0xAE (tentative)
1	PIV Mode Flags	Global behaviour flags (Section 3.10).	0x00–0xFF
2	MpSizeTotal	Total size of the complete configuration payload, least-significant byte first.	0x0000–0xFFFF
2	MpOffset	Offset of this fragment within the configuration payload, least-significant byte first.	0x0000–0xFFFF
2	MpFragmentSize	Length of the fragment payload, least-significant byte first.	0x0000–0xFFFF
1	Config Count	Number of configuration entries (1-32).	0x01–0x20
1	Global Interface Mask	Bitmask limiting interfaces available to all configurations (bit 0 contact, bit 1 contactless, bit 3 Bluetooth; bit 2 reserved and set to zero).	0x00–0x0B
4	Series Counter	Conditional field present only when PIV Auto is enabled. Least-significant byte first.	0x00000000–0xFFFFFFFF
4	Sequence Counter	Conditional field present only when PIV Auto is enabled. Least-significant byte first.	0x00000000–0xFFFFFFFF
32	Supplemental Entropy	Conditional fixed-width field present only when PIV Auto is enabled. All zeroes are permitted but not recommended.	32 octets
0-n	TLV Data	Sequence of Configuration Entry TLVs (Tag 0x01) that follow the Global Interface Mask, or follow the conditional PIV Auto fields when PIV Auto is enabled.	–

The *PIV Mode Flags* byte enables global behaviours such as autonomous operation and VCI usage (Table 25). Bits not listed remain reserved and **SHALL** be transmitted as zero.

Table 25: PIV Mode Flags

Mask	Description
0x01	Enable autonomous PIV (PIV Auto). The PD interrogates credentials, establishes VCI when possible, derives a KMAC256 challenge from the configured counters and entropy, and reports the signed response using <code>osdp_PIVSTATUSR</code> .
0x02	Permit VCI usage when available.
0x04	Enforce FICAM strict mode (PIV applications only).

Global and per-entry interface masks use the bit positions in Table 26; the Global Interface Mask bounds the interfaces a PD may use when processing all configuration entries.

Table 26: PIV Mode Interface Masks

Mask	Description
0x01	Permit ISO 7816 contact credential interface use.
0x02	Permit ISO 14443 contactless credential interface use.
0x04	Reserved and SHALL be transmitted as zero.
0x08	Permit Bluetooth credential interface use.
0x10–0x80	Reserved and SHALL be transmitted as zero.

After the flags, the payload uses standard multi-part framing fields and a configuration count (1-32). The count is followed by the required Global Interface Mask byte. If and only if PIV Auto is enabled in *PIV Mode Flags*, the Global Interface Mask is followed by the Series Counter, Sequence Counter, and fixed 32-byte Supplemental Entropy field before the TLV list. When PIV Auto is not enabled, those three conditional fields are absent and the TLV list begins immediately after the Global Interface Mask. Each configuration appears as a TLV:

- **Global Interface Mask (required)** — One-byte mask (no tag) that limits the interfaces the PD may use for all configurations (see Table 26). A value of 0x00 indicates no global restriction; reserved bits **SHALL** be zero.
- **Configuration Entry (Tag 0x01)** — Contains the data-model identifier, a per-entry interface mask, the AID length, and the optional AID:
 - *Data Model* (0x01 PIV, 0x02 TWIC, 0x03 CAC, values 0x04-0x7F reserved, 0x80-0xFF private use). CAC entries use the same interface and AID encoding as PIV and TWIC entries.
 - *Interface Mask* — Uses the same bit positions as the global mask (Table 26) and **SHALL NOT** assert interfaces excluded by the Global Interface Mask. A value of 0x00 means “any interface permitted by the global mask.”
 - *AID Length* (0-16) and *AID* bytes when provided. The legacy AID list referenced in earlier proposals is informational; it is not part of the official OSDP 2.2 specification.

PDs process configuration entries in order, applying the global mask (when present) before honouring each entry's mask. Invalid masks, unknown data-model identifiers, or malformed TLVs **SHALL** result in `osdp_PDERROR 0x0005`. Controllers **SHALL** send the command over a secure channel where required; the PD **SHALL** return `osdp_PDERROR 0x0006` if the necessary security conditions are not satisfied. Multi-part framing errors result in `osdp_PDERROR 0x0007`. If the PD cannot cache the requested configuration set, it **SHALL** respond with `osdp_PDERROR 0x1021`.

When PIV Auto is enabled, PDs interrogate the credential on presentation, establish VCI automatically when possible, derive the PIV Auto challenge, issue the PIV GENERAL AUTHENTICATE command, and report the signed response without additional ACU prompts. If a pairing code is required, the PD signals the state change via a solicited `osdp_CARDSTATUSR` (not via unsolicited poll); the ACU then provides the code using `osdp_PIV_XMITPAIRING`.

PIV Auto configuration **SHALL** be accepted only when OSDP Secure Channel 2.0 is active and the channel is not using a factory/default SCBK or other factory key material. A PD **SHALL** reject PIV Auto configuration or execution with `osdp_PDERROR 0x0006` when these preconditions are not met.

When bit 0x01 of *PIV Mode Flags* enables PIV Auto, the configuration **SHALL** specify a *Series Counter*, a *Sequence Counter*, and a 32-byte *Supplemental Entropy* field. The supplemental entropy field is fixed width and **SHALL** always be present; all zeroes are permitted for compatibility but are not recommended. The ACU **SHALL** ensure that a series counter value does not repeat for the same PD and that no sequence counter value is issued more than once within a series. One conforming implementation strategy is to maintain a monotonic series counter that increments at every ACU boot, similar to the SNMP engine boots counter, and a monotonic sequence counter for challenges issued within that boot series.

The PIV Auto challenge **SHALL** be derived from the active Secure Channel 2.0 session key material using two KMAC256 operations. First, the PD derives a 32-byte PIV Auto KDK with key material formed by concatenating S-ENC, S-MAC1, and S-MAC2; empty input data; output length 32 bytes; and customization string `OSDP-PIV-AUTO-KDK-v1`. Second, the PD derives the challenge digest using the PIV Auto KDK as the KMAC key, the fixed-width template as input data, output length L, and customization string `OSDP-PIV-AUTO-CHALLENGE-v1`. The fixed-width template **SHALL** contain the series counter, sequence counter, credential FASC-N, credential UUID, challenged PIV key reference, PIV algorithm identifier, and the 32-byte supplemental entropy field. A successful PIV Auto transaction **SHALL** include the credential UUID; if the UUID cannot be read, the PD **SHALL** report PIV Auto failure instead of deriving a success challenge with an all-zero UUID. The output length L is 32 bytes for SHA-256 profiles and 48 bytes for SHA-384 profiles. PIV Auto challenge derivation **SHALL NOT** use static factory keys.

3.11 PIV Auto Status Reply (`osdp_PIVSTATUSR`)

After a PIV Auto card challenge completes, the PD reports the result as a poll response using reply code 0x89 (tentative). The first `osdp_POLL` response after completion **SHALL** contain `osdp_PIVSTATUSR`; subsequent fragments, when required, **SHALL** use the same reply and advance *MpOffset* until the complete status payload is delivered.

Table 27: PIV Auto Status Reply (osdp_PIVSTATUSR)

Size	Name	Meaning	Value
1	CMND	Reply identifier.	0x89 (tentative)
2	MpSizeTotal	Total size of the complete PIV Auto status payload, least-significant byte first.	0x0000-0xFFFF
2	MpOffset	Offset of this fragment within the complete status payload, least-significant byte first.	0x0000-0xFFFF
2	MpFragmentSize	Length of the status payload bytes in this fragment, least-significant byte first.	0x0000-0xFFFF
1	Result	PIV Auto result for the challenged credential. See Table 28.	0x00-0xFF
1	Detail	Additional result detail. Reserved as 0x00 when Result is 0x00; may carry SW1 SW2 low byte when Result is 0x06.	0x00-0xFF
4	Series Counter	Series counter value used to derive the challenge, least-significant byte first.	0x00000000-0xFFFFFFFF
4	Sequence Counter	Sequence counter value used to derive the challenge, least-significant byte first.	0x00000000-0xFFFFFFFF
25	FASC-N	FASC-N read from the credential.	200-bit FASC-N
16	UUID	Credential UUID read from the credential. Success replies SHALL carry a nonzero UUID; failure replies may transmit all zeroes when the UUID is unavailable.	16 octets
1	Key Reference	PIV key reference challenged by the PD.	For example 0x9E
1	Algorithm Identifier	PIV GENERAL AUTHENTICATE algorithm identifier.	For example 0x07
2	Signed Response Length	Total length of the signed response, least-significant byte first.	0x0000-0xFFFF
0-n	Signed Response Fragment	Contiguous bytes of the signed response carried in this fragment.	Variable

Result and detail codes follow Table 28. When *Result* is 0x00, *Detail* is reserved and **SHALL** be 0x00. When *Result* is not 0x00, the PD **SHALL** set *Signed Response Length* to 0x0000 unless a credential response was received and is useful for ACU diagnostics.

Table 28: PIV Auto Status Result Codes

Value	Description
0x00	Success. <i>Detail</i> is 0x00, UUID is nonzero, and Signed Response carries the complete returned Dynamic Authentication Template.
0x01	Credential not present or removed before PIV Auto completed.
0x02	VCI establishment or required VCI validation failed.
0x03	Required credential data object, FASC-N, UUID, key, or certificate evidence is unavailable.
0x04	Requested PIV key reference or algorithm is not supported for PIV Auto.
0x05	Credential security status is not satisfied, including missing PIN or pairing preconditions.
0x06	Credential GENERAL AUTHENTICATE operation failed; <i>Detail</i> may carry SW1 SW2 when available.
0x07	Signed response verification failed.
0x08	Timeout or PD internal failure before a definitive credential result was available.
0x09–0x7F	Reserved for future standard result codes.
0x80–0xFF	Private use.

The reply reports the exact series and sequence counters used for the challenge, the FASC-N and UUID read from the credential, and the signed response returned by the credential. The ACU uses these fields to verify counter uniqueness, bind the response to the credential identity, and validate the signature and certificate path using ACU trust policy. A success result **SHALL** include a nonzero credential UUID. The PD **SHALL** preserve the credential's signed response bytes as returned by the PIV operation, excluding transport status words such as SW1/SW2.

If the complete signed response does not fit in one OSDP response, the PD **SHALL** fragment `osdp_PIVSTATUSR` using the OSDP 2.2 multi-part convention. *MpSizeTotal* is the total status payload size, *MpOffset* is the zero-based offset of the fragment, and *MpFragmentSize* is the number of payload bytes carried in that fragment. The fixed fields appear at offset zero; later fragments carry the next contiguous bytes of the signed response.

3.12 VCI Trust Anchor Load (`osdp_PIV_VCILOADTA`)

The `osdp_PIV_VCILOADTA` command (command code 0xAF, tentative) loads, replaces, or deletes a Virtual Contact Interface (VCI) trust anchor. The ACU **SHALL** validate and authorize each trust anchor

under its deployment certificate policy before loading it. Trust anchors must be cached before a PD can verify card certificates during VCI establishment.

Table 29: VCI Trust Anchor Load (osdp_PIV_VCILOADTA)

Size	Name	Meaning	Value
1	CMND	Command identifier.	0xAF (tentative)
1	Operation	0x00 load/replace anchor; 0x01 delete anchor. Anchor inventory is requested with osdp_PIV_VCITASTATUS.	0x00–0x01
1	Trust Anchor ID	Controller-assigned identifier for the anchor (1-128).	0x00–0xFF
8	Issuer Identifier Number	The leftmost eight bytes of the subjectKeyIdentifier from the content-signing X.509 certificate. This value SHALL match the IIN field in the VCI trust anchor record (tag 0x42) and is used as the anchor selector during VCI establishment.	0x00–0xFF
2	Anchor Length	Total size of the variable-length VCI trust anchor record, least-significant byte first. Set to 0x0000 when deleting.	0x0000–0x0240
2	MpSizeTotal	Total size of the complete anchor data, least-significant byte first. For load operations, this value equals AnchorLength. Set to 0x0000 when deleting.	0x0000–0x0240
2	MpOffset	Offset of this fragment within the anchor data, least-significant byte first. Set to 0x0000 when deleting.	0x0000–0x023F
2	MpFragmentSize	Length of the data block in this fragment, least-significant byte first. MpOffset + MpFragmentSize SHALL NOT exceed AnchorLength. Set to 0x0000 when deleting.	0x0000–0x0240
0-n	Anchor Data	Fragment bytes of the OSDP-defined VCI trust anchor record. Omitted for delete operations.	0x00–0xFF

The metadata block (operation, identifier, IIN, anchor length) appears only in the first fragment where *MpOffset* = 0; subsequent fragments carry additional anchor bytes. Controllers set *Operation* to 0x00 to load or replace the anchor identified by *Trust Anchor ID*. Set *Operation* to 0x01 with *Anchor Length* 0x0000 to delete the anchor. Inventory is requested with *osdp_PIV_VCITASTATUS*, not with *osdp_PIV_VCILOADTA*. The *Issuer Identifier Number (IIN)* is an OSDP-defined anchor selector: it **SHALL** be the leftmost eight bytes of the subjectKeyIdentifier from the content-signing X.509 certificate associated with this anchor. This value **SHALL** match the IIN field (tag 0x42) in the VCI trust anchor record.

The *Anchor Length* declares the exact total size of the variable-length VCI trust anchor record. For load operations, *MpSizeTotal* **SHALL** equal *Anchor Length*, the sum of all accepted fragment lengths **SHALL** equal *Anchor Length*, and no fragment **SHALL** extend beyond *Anchor Length*. A conforming VCI trust anchor record **SHALL NOT** exceed 576 bytes (0x0240). The PD **SHALL** accept any valid record length from the minimum BER-TLV encoding through 0x0240, subject to supported-key and storage-capacity checks.

PD storage accounting for VCI trust anchors is variable-length. A PD **SHALL NOT** treat every anchor as consuming the maximum 0x0240 bytes unless that is the PD's documented physical allocation model. When reporting capacity, loading, replacing, or deleting anchors, the PD **SHALL** account for stored anchors by their actual *Anchor Length* values and **SHALL** use the same accounting rule for both *osdp_PIV_VCITASTATUSR Available Bytes* and *osdp_PIV_VCILOADTA* storage acceptance. A replacement operation **SHALL** be evaluated after releasing the storage occupied by the existing anchor with the same *Trust Anchor ID*; if the replacement cannot be completed, the prior anchor **SHALL** remain available. A delete operation **SHALL** release the storage associated with the deleted anchor.

Table 30: VCI Trust Anchor Record Format

Tag	Length	Name	Value
0x7F50	Variable	VCI Trust Anchor Record	OSDP-defined record used to verify a secure messaging CVC or an Intermediate CVC.
0x80	1	Record Profile Identifier	0x01.
0x42	8	Issuer Identifier Number	Anchor selector. This value SHALL match the IIN in the <i>osdp_PIV_VCILOADTA</i> metadata.
0x7F49	1-550	SubjectPublicKeyInfo	DER-encoded X.509 SubjectPublicKeyInfo for the public key trusted by the ACU. Supported public-key algorithms are RSA public keys up to 4096 bits and EC public keys for P-256 or P-384.

The VCI trust anchor record is intentionally not a NIST SP 800-73 Intermediate CVC and does not carry X.509 path-validation state. The ACU is responsible for selecting, loading, unloading, and validating the VCI anchors it permits a PD to use before the PD relies on them. The PD uses the record only as a local public-key verifier selected by IIN. The maximum record size is derived from the largest supported key case: a 550-byte RSA-4096 SubjectPublicKeyInfo plus the fixed fields above and BER-TLV length overhead.

The resulting worst case is 573 bytes; this specification rounds the maximum accepted record size to 576 bytes.

Use during VCI establishment When the PD receives the secure messaging CVC from the credential, it **SHALL** first extract the Issuer Identification Number from tag 0x42. If a loaded VCI trust anchor has a matching IIN, the PD **SHALL** use that anchor's SubjectPublicKeyInfo to verify the secure messaging CVC signature. If verification succeeds, the PD may continue VCI establishment using the public key in the secure messaging CVC.

If no loaded VCI trust anchor matches the secure messaging CVC IIN, the PD **SHALL** retrieve the Secure Messaging Certificate Signer data object (PIV data object 0x1017, tag 5FC122) from the credential and inspect it for one Intermediate CVC (tag 0x7F21). If the object is absent or no Intermediate CVC is present, validation fails. If an Intermediate CVC is present, the PD **SHALL** verify all of the following before continuing:

- The secure messaging CVC IIN (tag 0x42) equals the Subject Identifier (tag 0x5F20) in the Intermediate CVC.
- The Intermediate CVC Role Identifier (tag 0x5F4C) is 0x12.
- A loaded VCI trust anchor exists whose IIN equals the Intermediate CVC IIN (tag 0x42).
- The Intermediate CVC signature verifies using the matching VCI trust anchor public key.
- The secure messaging CVC signature verifies using the public key carried in the Intermediate CVC.

The PD validates only the CVC chain needed for VCI establishment: either the loaded trust anchor directly verifies the secure messaging CVC, or the loaded trust anchor verifies the Intermediate CVC that then verifies the secure messaging CVC. The PD **SHALL NOT** establish VCI unless this CVC validation succeeds. The PD **SHALL NOT** perform full X.509 path validation, validity-period evaluation, certificate-policy evaluation, or revocation processing for VCI trust anchors; those checks are completed by the ACU before loading anchors and again as an acceptance policy after VCI state is reported.

ACU post-VCI validation After receiving `osdp_CARDSTATUSR` with VCI established, the ACU **SHALL** validate the content-signing certificate corresponding to the reported *VCI Anchor IIN* before treating the VCI session as accepted for the deployment. This validation is an additional ACU acceptance check and does not replace the PD's required CVC validation during VCI establishment. The ACU **SHALL** perform all of the following:

- Obtain the content-signing X.509 certificate. If the ACU does not already possess a cached copy whose subjectKeyIdentifier leftmost-eight-bytes equals the *VCI Anchor IIN*, the ACU **SHALL** use `osdp_PIVDATA` to retrieve the Secure Messaging Certificate Signer data object (PIV data object 0x1017, tag 5FC122) and extract the certificate from inner tag 0x70.

- Verify that the leftmost eight bytes of the certificate's subjectKeyIdentifier equal the *VCI Anchor IIN* reported in `osdp_CARDSTATUSR`.
- Verify that the certificate's SubjectPublicKeyInfo matches the SubjectPublicKeyInfo stored in the loaded VCI trust anchor record for that IIN.
- Verify that the certificate includes the PIV content-signing extended key usage OID (2.16.840.1.101.3.6.7) or the PIV-I equivalent (2.16.840.1.101.3.8.7).
- Verify that the current time falls within the certificate's validity period (`notBefore` $\leq$ `now` $\leq$ `notAfter`). If the certificate has expired or is not yet valid, the ACU **SHALL** reject the VCI session and **SHALL** unload the corresponding trust anchor.
- Perform X.509 path validation and revocation checking in accordance with the deployment's certificate validation policy.

Table 31: VCI Trust Anchor Examples from NIST Vectors

Vector	Content Signer	Anchor IIN	SPKI	Record
card_2, card_5	Test PIV Content Signer 4 (EC P-384, direct)	B6103792270FBD08	120 bytes	140 bytes
card_3	Test PIV Content Signer 1 (RSA-2048, intermediate)	9BC5E7735F14EC7F	294 bytes	317 bytes
card_4	Test PIV Content Signer 3 (EC P-256, direct)	1905976BDDCA823D	91 bytes	110 bytes
card_16	Test PIV-I Content Signer 1 (RSA-2048, intermediate)	4CC9C7B3A4F1C83E	294 bytes	317 bytes

In the `card_16` intermediate example, the secure messaging CVC IIN is 6F8FF48F42B3E22E; the Intermediate CVC Subject Identifier (tag 0x5F20) is also 6F8FF48F42B3E22E; and the Intermediate CVC IIN (tag 0x42) is 4CC9C7B3A4F1C83E, matching the loaded RSA trust anchor.

PDs **SHALL** reconstruct the anchor from the fragments before storing it. The PD **SHALL** verify that the reconstructed byte string is exactly *Anchor Length* bytes and is a single 0x7F50 record before committing it to storage. A successful operation returns `osdp_ACK`. Failures return:

- `osdp_NAK` function-not-supported when the PD does not support trust-anchor caching.
- `osdp_PDERROR 0x0005` for malformed metadata (unknown operation, reserved bits, invalid lengths, `MpSizeTotal` not equal to *Anchor Length*, or a fragment range outside *Anchor Length*).
- `osdp_PDERROR 0x0007` when multi-part sequencing fails.
- `osdp_PDERROR 0x1021` when the PD lacks storage space for the anchor.

PDs are not required to maintain a real-time clock. PDs may retain either the supplied VCI trust anchor record or an equivalent internal representation so long as the public key and supplied IIN remain available for the verification flow above.

Trust anchors may be stored in volatile or non-volatile memory at the PD's discretion. Controllers should not assume retention beyond the device's documented behaviour.

3.13 VCI Trust Anchor Status (osdp_PIV_VCITASTATUS)

The osdp_PIV_VCITASTATUS command (command code 0xA6, tentative) enumerates cached trust anchors and reports remaining storage capacity. The request carries no payload. The reply uses the standard multi-part fields (TOTAL, OFFSET, DATA_LEN) so PDs can stream the inventory across one or more fragments using reply code 0x88 (tentative).

Table 32: VCI Trust Anchor Status Request (osdp_PIV_VCITASTATUS)

Size	Name	Meaning	Value
1	CMND	Command identifier.	0xA6 (tentative)

Table 33: VCI Trust Anchor Status Reply (osdp_PIV_VCITASTATUSR)

Size	Name	Meaning	Value
1	CMND	Reply identifier.	0x88 (tentative)
2	TOTAL	Length of the complete status payload, least-significant byte first.	0x0000-0xFFFF
2	OFFSET	Offset of this fragment within the status payload, least-significant byte first.	0x0000-0xFFFF
2	DATA_LEN	Length of the data block in this fragment, least-significant byte first.	0x0000-0xFFFF
1	Total Anchor Count	Number of cached trust anchors.	0x00-0xFF
2	Available Bytes	Remaining storage capacity for additional variable-length trust anchor records, least-significant byte first.	0x0000-0xFFFF

Size	Name	Meaning	Value
0-n	Anchor Records	Zero or more 12-byte inventory records. Each record contains: TrustAnchorID (1 byte for load/delete management), anchor selector IIN (8 bytes from the stored trust anchor record, not necessarily the secure CVC IIN), AnchorLength (2 bytes LSB first, maximum 0x0240), and Attributes (1 byte, reserved). Records that span fragments continue in subsequent replies.	-

The PD first reports the total number of anchors and the remaining free bytes. *Available Bytes* is the amount of trust-anchor storage that remains for additional variable-length VCI trust anchor record data after accounting for currently stored anchors by their actual *Anchor Length* values. Each record thereafter lists a trust anchor by management ID, anchor selector IIN, anchor length, and reserved attributes (transmitted as zero). The IIN in this inventory is the IIN from the stored VCI trust anchor record and is not necessarily the secure messaging CVC IIN reported in `osdp_CARDSTATUSR`. Error responses follow the standard pattern: `osdp_NAK` function-not-supported when the command is not implemented, and `osdp_PDERROR 0x0007` for multi-part errors.

Table 34: Trust Anchor Record Structure

Field	Size	Meaning	Value Range
Trust Anchor ID	1 byte	Identifier assigned by the ACU; reused to replace or delete an anchor.	0x00-0xFF
Issuer Identifier Number (IIN)	8 bytes	Anchor selector supplied in the VCI trust anchor record.	0x00-0xFF
Anchor Length	2 bytes (LSB first)	Total size of the stored variable-length VCI trust anchor record supplied for this anchor.	0x0000-0x0240
Attributes	1 byte	Reserved for future use; transmitted as zero.	0x00

A References

A.1 Normative References

- Security Industry Association, *Open Supervised Device Protocol (OSDP) v2.2*, 2020.
- National Institute of Standards and Technology, *NIST Special Publication 800-73-4, Interfaces for Personal Identity Verification (PIV) — Part 2: PIV Card Application Card Command Interface*, February 2015.
- National Institute of Standards and Technology, *NIST Special Publication 800-73-5, Interfaces for Personal Identity Verification (PIV) — Part 2: PIV Card Application Card Command Interface*, 2024.
- National Institute of Standards and Technology, *NIST Special Publication 800-78-5, Cryptographic Algorithms and Key Sizes for Personal Identity Verification*, 2024.
- National Institute of Standards and Technology, *Federal Information Processing Standards Publication 201-3, Personal Identity Verification (PIV) of Federal Employees and Contractors*, January 2022.
- National Institute of Standards and Technology, *NIST Special Publication 800-185, SHA-3 Derived Functions: cSHAKE, KMAC, TupleHash, and ParallelHash*, December 2016.
- International Organization for Standardization and International Electrotechnical Commission, *ISO/IEC 7816-4, Identification cards — Integrated circuit cards — Part 4: Organization, security and commands for interchange*.
- Internet Engineering Task Force, *RFC 5280, Internet X.509 Public Key Infrastructure Certificate and Certificate Revocation List (CRL) Profile*, May 2008.
- Internet Engineering Task Force, *RFC 2119, Key words for use in RFCs to Indicate Requirement Levels*, March 1997; and *RFC 8174, Ambiguity of Uppercase vs Lowercase in RFC 2119 Key Words*, May 2017.

A.2 Informative References

- National Institute of Standards and Technology, *NIST Internal Report 8347, Personal Identity Verification Cards, Version 2*, 2021.
- U.S. General Services Administration, *Federal Identity, Credential, and Access Management (FICAM) Architecture*, Revision 3, 2023.
- Open Physical, *High Speed Autonomous PIV: Concept Paper*, 2024.

B Sample osdp_PIVDATA Requests (Informative)

This appendix illustrates representative requests that conforming ACUs may issue using the clarified osdp_PIVDATA command. Object identifiers align with NIST SP 800-73-4 Part 2, Table 6. Each example shows the bytes that appear in the OSDP payload after the CMND field; the PD strips leading zeroes before constructing the ISO 7816-4 APDU.

Cardholder Unique Identifier (CHUID) Object identifier 0x5FC102, tag 0x00, offset 0x0000. The PD transmits GET DATA for tag 0x5FC102 and returns the complete CHUID BER-TLV so the ACU can validate the FASC-N, GUID, and expiration fields.

Card Capability Container (CCC) Object identifier 0x5FC107, tag 0x00, offset 0x0000. Use this request during device discovery to read feature flags (contact/contactless support, optional biometrics) prior to issuing PIV Mode updates.

Discovery Object Object identifier 0x00007E, tag 0x00, offset 0x0000. The PD forwards tag 0x7E with its nested data so the ACU can parse the application identifier (tag 0x4F) and PIN Usage Policy (tag 0x5F2F). To retrieve only the AID, repeat the command with tag 0x4F.

Biometric Information Templates (BIT) Group Template Object identifier 0x007F61, tag 0x00, offset 0x0000. Large responses for fingerprint or iris templates require multiple osdp_PIVDATAR fragments; the TOTAL/OFFSET/DATA_LEN fields support reconstruction.

Selective Tag Retrieval Object identifier 0x00007E, tag 0x4F, offset 0x0000. The PD returns the complete PIV Application Identifier BER-TLV nested inside the Discovery Object, including tag 0x4F, length, and value. Tags that occupy more than one byte (for example 0x7F61) are outside the scope of this selector and must be obtained by requesting tag 0x00.

C High-Speed Autonomous PIV Data Templates (Informative)

This appendix records the PIV Auto challenge, ISO/IEC 7816 GENERAL AUTHENTICATE templates, and fixture sources used by the demonstration vectors. The APDU examples are validated by the repository tests against the same helper code used by the simulator and replay tools.

C.1 Vector Sources

Table 35: PIV Auto Vector Sources

Source	Use	Notes
NIST SD 33 / IR 8347	Live-card APDU fixtures	Cards 2, 3, 4, 5, and 16 provide VCI contactless enrolment, OPACITY, and key-challenge transcripts.
NIST SM/VCI captures	CVC and trust anchors	Provides validator-ready 7F21, 5FC122, and 7F50 objects.
GSA FICAM card-builder	Local keys and certificates	Used only as reusable local test material for source-backed simulation.
Generated YubiKey PIV	Missing algorithm profiles	Reusable RSA-1024, RSA-3072, and P-384 test keys are committed as public fixtures and loaded through <code>ykman piv</code> ; RSA-1024 is informative legacy material only.

C.2 Supported Profiles

Table 36: PIV Auto Supported Profiles

Profile	Alg	Key Ref	PIV Auto	Coverage
RSA-1024	06	9A, 9E	No	Informative legacy/non-current test-only material.
RSA-2048	07	9A, 9E	Yes	NIST SD 33 cards 2, 3, 16; generated-card; GSA-backed simulation.
RSA-3072	05	9A, 9E	Yes	Committed YubiKey material; requires YubiKey firmware with RSA-3072 support.
ECC P-256	11	9A, 9E	Yes	NIST SD 33 cards 4 and 5; generated-card coverage.
ECC P-384	14	9A, 9E	Yes	Committed YubiKey material. NIST card 5 also provides 9C/9D APDU examples.

C.3 Fixed-Width KMAC Challenge Template

Table 37: PIV Auto KMAC Challenge Template

Field	Size	Encoding	Notes
Template label	14	ASCII plus version byte	OSDP-PIV-AUTO followed by 01.
Series Counter	4	LSB first	ACU-managed boot or epoch counter, for example 07000000.
Sequence Counter	4	LSB first	ACU-managed per-series counter, for example 2A000000.
FASC-N	25	Raw bytes	Credential FASC-N from CHUID.
UUID	16	Raw bytes	Credential UUID. Successful PIV Auto examples use the credential UUID; unavailable UUID is a failure condition.
PIV Key Reference	1	Raw byte	Challenged key reference, normally 9A or 9E.
PIV Algorithm	1	Raw byte	Algorithm identifier: 05, 06, 07, 11, or 14.
Supplemental Entropy	32	Raw bytes	Fixed-width ACU entropy field; all zeroes are valid for test vectors only.

The challenge digest is produced in two steps. First, derive PIV-AUTO-KDK=KMAC256(key=S-ENC|S-MAC1||S-MAC2,data=empty,mac_len=32,custom="OSDP-PIV-AUTO-KDK-v1"). Second, derive the challenge as KMAC256(key=PIV-AUTO-KDK,data=fixed-widthtemplate,mac_len=L,custom="OSDP-PIV-AUTO-CHALLENGE-v1"). L is 32 bytes for SHA-256 profiles and 48 bytes for SHA-384 profiles. The supplemental entropy field is always present and fixed at 32 bytes; all zeroes are valid for test vectors but not recommended for deployments.

C.4 Dynamic Authentication Templates

C.4.1 RSA Key Challenge Input

Table 38: RSA Dynamic Authentication Template

Tag / Bytes	Length	Name	Value
7C	Variable	Dynamic Authentication Template	Outer template for the PIV GENERAL AUTHENTICATE command data.
82	00	Response request	Empty value requesting the card response in tag 82.
81	Modulus length	Challenge block	EMSA-PKCS1-v1_5 encoded digest block with the same length as the RSA modulus.
0001FF...FF00	Variable	Padding prefix	EMSA-PKCS1-v1_5 block prefix before the digest information.
30...	Variable	DigestInfo	ASN.1 DigestInfo for SHA-256 or SHA-384 followed by the KMAC-derived digest; SHA-256 uses prefix 3031300D060960864801650304020105000420 and SHA-384 uses prefix 3041300D060960864801650304020205000430.

Table 39: RSA Dynamic Authentication Examples

Profile	Alg	Key Ref	Tag 81	Response 82	Command APDU
RSA-1024	06	9A/9E	128 bytes	128 bytes	Informative legacy/non-current test-only APDU.
RSA-2048	07	9A/9E	256 bytes	256 bytes	274-byte extended APDU.
RSA-3072	05	9A/9E	384 bytes	384 bytes	402-byte extended APDU.

C.4.2 ECC Signature Input

Table 40: ECC Dynamic Authentication Template

Tag / Bytes	Length	Name	Value
7C	Variable	Dynamic Authentication Template	Outer template for the PIV GENERAL AUTHENTICATE command data.
82	00	Response request	Empty value requesting the card response in tag 82.
81	Digest length	Challenge digest	KMAC-derived digest supplied directly to the ECC signing operation.
Digest bytes	32 or 48 bytes	Digest value	SHA-256-width digest for P-256; SHA-384-width digest for P-384.

Table 41: ECC Dynamic Authentication Examples

Profile	Alg	Key Ref	Tag 81	Response 82	Command APDU
ECC P-256	11	9A/9E	32 bytes	DER ECDSA signature, about 70-72 bytes	44-byte short APDU.
ECC P-384	14	9A/9E	48 bytes	DER ECDSA signature, about 102-104 bytes	60-byte short APDU.

C.4.3 Worked Dynamic Authentication Examples

Worked examples are generated from
 test-vectors/piv-auto-demo/generated/piv-auto-simulation-report.json.

Table 42: PIV Auto Worked Example: PIV Authentication RSA-1024

Field	Value
Profile	9a-rsa1024
Source	Committed YubiKey loader material in test-vectors/piv-auto-yubikey-material/9a-rsa1024. Informative legacy/non-current test-only coverage.
Algorithm ID	0x06
Key Reference	0x9A
KMAC Challenge Digest	5B2FC08860E3999EEF8F6CF5D2B0C9AE05CBB28A5B423BFE4F646F6F47FCA4F2

Table 45: PIV Auto Worked Example: Card Authentication RSA-2048

Field	Value
Profile	9e-rsa2048
Source	Committed RSA-2048 card-authentication key in test-vectors/piv-auto-demo/source-p12, originally imported from GSA FICAM card-builder. NIST SD 33 / IR 8347 and generated-card coverage.
Algorithm ID	0x07
Key Reference	0x9E
KMAC Challenge Digest	0864C776F2374124D3E63F0B0B29FC1C5F0E8FF8BB1FA80E2723293B86A0158E
Tag 81 Challenge Input	0001FF FF FF FF FF FF FF FF FF FF 003031300D060960864801650304020105000420 0864C776F2374124D3E63F0B0B29FC1C5F0E8FF8BB1FA80E2723293B86A0158E
GENERAL AUTHENTICATE APDU	0087079E00010A7C8201068200818201000001FFFFFFFFFFFFFFFFFFFFFFFFFFFF FF FF FF FF FF FF FF FF FF 003031300D0609608648016503040201050004200864C776F2374124D3E63F0B0B29FC1C5F0E8FF8BB1FA80E2723293B86A0158E00
Dynamic Authentication Response	7C820104828201009BB1AA3875DB4E41107DD91CEAAC36FE4CE5F969E1F7B9B1 99DFAB38CFA1EE36FBF197CD2777C4A6CE082623C9F36A9DC2514FE54DD12934 AFCAE16DFE14FFCF0F58E0A0F46567444968F1DE78B496B4222DB913A87B850 139B1E2A730124C4ED169919F1EB5FA3E15617E89BB9CED8FA996A63E126C239 EB6D142A5746FBEC3DFA3B287F944EE45586BA5AEB64087BDA0661DF7254C42D 9768141F7243382D61CC43A8CDBE4ADA849E05F013D947DF7FA608C38ABA3841 B038AA69D77461B00053F59C4D34906BFBD470A277D0722AAECD83117DE906B FCD36A16EFB00B3EACD4414F48528CCA4BE57826B4312424A65DABAD6FA1ECA3 DAF55F6E6E1022C89000
Response Tag 82 Length	256 bytes

Table 46: PIV Auto Worked Example: PIV Authentication RSA-3072

[illegible]

Field	Value
GENERAL AUTHENTICATE APDU	<pre> 0087059E00018A7C8201868200818201800001FFFFFFFFFFFFFFFFFFFFFFFF FF FF FF FF FF FF FF FF FF FF FF FF FF FF 003041300D0609608648016503040202050004 304C98712B25DB7F477096FA3D0AC93FA2CA6BBE89CB291AC26EDF0D8312CA61 DOCA238F1656247B2DF38E8B72D9AF7F5E00 </pre>
Dynamic Authentication Response	<pre> 7C8201848282018074FA8827F5C2CFDAA5718679727015CD5A928828997F2F14 23D033DE8AADEB3E86431170A714F5155B830F65F3F854120B16BC50F701C0EE A0278DB7EC2D7D6D587BEDB32F5E34EF1421E2C42543FF7DEEF7C5A41A01275B 0BC126AF1208BBBD1E2C65149BC74C9E2CD00D822081B22E5CCB5090D2BD3970 CA309D2124D79E9A28978D5DDF26E3FA8E37C360521C5BBE1AB27FACC24FF4C5 9AB46B7D571777B22E3E4981593D9F689DA3D33F3DE7C385C4315A7F154BE461 968EA05A3315C4424717DD6F3CBF7FB94A135AE33573E58A18185F7E25771E6C 4AB2B89D50007C7A60E6D65CF3BE58F8901B25B867D0FF97981BCBEE2BFCF5E5 7CEB5A81F40EEB44E52A0BD7F538F932AB4F1732828F2E95C4F203967ADF5983 A285A127E6C0D7C2048261290B6B58D8110056DFE65547A7C1BDF78C6FEA62A8 E07EC932AC6B58AD13C9423AAD5815C5251BCECEBBDB181F0B1C362163373463 5FD48CD1AD16798C9B67CDF2BAB290DE0EA0BC5BA54E41736BA4885AF72D729A 5168DC09DDD4B6629000 </pre>
Response Tag 82 Length	384 bytes

Table 48: PIV Auto Worked Example: PIV Authentication ECC P-256

Field	Value
Profile	9a-ecp256
Source	Committed P-256 signing key in test-vectors/piv-auto-demo/source-p12, originally imported from GSA FICAM card-builder. NIST SD 33 / IR 8347 and generated-card coverage.
Algorithm ID	0x11
Key Reference	0x9A
KMAC Challenge Digest	A3C81D3205FFB28A2AAAB0EDC21E8B3BEF5BD5F685FF5C686502CA5609B45851
Tag 81 Challenge Input	A3C81D3205FFB28A2AAAB0EDC21E8B3BEF5BD5F685FF5C686502CA5609B45851
GENERAL AUTHENTICATE APDU	0087119A267C2482008120A3C81D3205FFB28A2AAAB0EDC21E8B3BEF5BD5F685FF5C686502CA5609B4585100

Field	Value
Dynamic Authentication Response	7C4A824830460221009D25A44E56572983819019DF26A2720E462A11D2CB3CD86A9142DEF4BCC1271B022100D084982BB9D3E2D3E6B0965956C71DBA0FE44FE328C0A2BD72951DFB4D9296FE9000
Response Tag 82 Length	72 bytes

Table 49: PIV Auto Worked Example: Card Authentication ECC P-256

Field	Value
Profile	9e-ecp256
Source	Committed P-256 signing key in test-vectors/piv-auto-demo/source-p12, originally imported from GSA FICAM card-builder. NIST SD 33 / IR 8347 and generated-card coverage.
Algorithm ID	0x11
Key Reference	0x9E
KMAC Challenge Digest	965AF088B02DF65E7C299ABF53A5AB2DCBB3C789458DE8D0C819B19A9266658D
Tag 81 Challenge Input	965AF088B02DF65E7C299ABF53A5AB2DCBB3C789458DE8D0C819B19A9266658D
GENERAL AUTHENTICATE APDU	0087119E267C2482008120965AF088B02DF65E7C299ABF53A5AB2DCBB3C789458DE8D0C819B19A9266658D00
Dynamic Authentication Response	7C4A824830460221008B204FA48FAA27A57D6E90F985199F284D2A8E0D0F2E051BBFBCACBA824619B9022100C6E04880D322DE7D10823EC78C8EA4CCB69FE394F694F1D12C797FA7BCAD76959000
Response Tag 82 Length	72 bytes

Table 50: PIV Auto Worked Example: PIV Authentication ECC P-384

Field	Value
Profile	9a-ecp384
Source	Committed YubiKey loader material in test-vectors/piv-auto-yubikey-material/9a-ecp384. Generated-card coverage.
Algorithm ID	0x14
Key Reference	0x9A
KMAC Challenge Digest	C813683447986D513FC17CED5DB4D335BFBEAFB1AF9E5200C6B7B1C1E3535D700C2E05F9011717E153998832B7588D28

Field	Value
Tag 81 Challenge Input	C813683447986D513FC17CED5DB4D335BFBEAFB1AF9E5200C6B7B1C1E3535D700C2E05F9011717E153998832B7588D28
GENERAL AUTHENTICATE APDU	0087149A367C3482008130C813683447986D513FC17CED5DB4D335BFBEAFB1AF9E5200C6B7B1C1E3535D700C2E05F9011717E153998832B7588D2800
Dynamic Authentication Response	7C6A82683066023100D88ABABB9893D88341CFA2F1AB6E3D35E4AF02F4F8D085B5A3DE63574DCF694741D869066B8E6297429120614DCC9CE40231009A3F203BB5EA98839008B0D25F98C98DE8E87622F61279B2F2B29343912999A36311102B310F9985844BCC248350377B9000
Response Tag 82 Length	104 bytes

Table 51: PIV Auto Worked Example: Card Authentication ECC P-384

Field	Value
Profile	9e-ecp384
Source	Committed YubiKey loader material in test-vectors/piv-auto-yubikey-material/9e-ecp384. Generated-card coverage.
Algorithm ID	0x14
Key Reference	0x9E
KMAC Challenge Digest	F3480C6C1DAD008D3E14D0C815D381F01420E9D3402A175BED097C6949E4BA447FA0A11745CE4D054A95B10A9D5CC689
Tag 81 Challenge Input	F3480C6C1DAD008D3E14D0C815D381F01420E9D3402A175BED097C6949E4BA447FA0A11745CE4D054A95B10A9D5CC689
GENERAL AUTHENTICATE APDU	0087149E367C3482008130F3480C6C1DAD008D3E14D0C815D381F01420E9D3402A175BED097C6949E4BA447FA0A11745CE4D054A95B10A9D5CC68900
Dynamic Authentication Response	7C6982673065023034C6B2659FBE26A78D2615F8C0EBF2E9674292D9DE3CE8A31D9DA297B5AE236AC886E9E9C288D42474564BF0C46AE40C023100CCBD59A047D73F9FD58F60218FBCB4F99B18EBD7EB70EFE5CF64D5EB1A65F1676A867E1B1E5EB4CF9F34B44BE905D29C9000
Response Tag 82 Length	103 bytes

C.5 APDUs

Table 52: PIV Auto APDU Reference: Card Preparation

Operation	Command APDU	Success Response	Purpose
Select PIV Application	00A404000BA000000308000010000100	9000	Selects the PIV application before reading objects or authenticating.
Get Discovery Object	00CB3FFF035C017E	7E...9000	Reads the Discovery Object so the PD can determine application and PIN/VCI policy.
Verify PIN (optional)	0020008008313233343536FFFF	9000	Verifies example PIN 123456 padded with FF; failure responses include 63CX and 6983.

Table 53: PIV Auto APDU Reference: GENERAL AUTHENTICATE Command

Field	Encoding	Source	Notes
CLA	00	ISO/IEC 7816-4	Use 0C when a credential secure channel is active.
INS	87	ISO/IEC 7816-4	GENERAL AUTHENTICATE instruction.
P1	Algorithm ID	PIV profile	05, 07, 11, or 14 for supported PIV Auto profiles; 06 appears only in informative RSA-1024 legacy examples.
P2	Key Reference	PIV profile	9A or 9E for PIV Auto.
Lc	Short or extended	Command length	Short APDUs use one length byte; extended APDUs use 00 followed by two length bytes.
Command data	7C...820081.....	Dynamic Authentication Template	Tag 82 is empty to request a response; tag 81 carries the exact challenge input.
Le	00	Response request	Requests the maximum response supported by the card interface.

Table 54: PIV Auto APDU Reference: GENERAL AUTHENTICATE Response

Case	Response APDU	PD Handling	OSDP Payload
Complete response	7C...82...9000	Strip only the ISO status word.	Carry the complete 7C Dynamic Authentication Template in osdp_CRAUTHR or osdp_PIVSTATUSR.
Additional response bytes	61XX	Issue GET RESPONSE with Le X X.	Concatenate returned data before reporting the final 7C template.
GET RESPONSE command	00C00000XX	Retrieve remaining card response bytes.	Not exposed directly to the ACU.
Credential error	69.. or 6A..	Preserve command failure detail.	Report via osdp_PDERRORR when available; do not treat as a valid signed response.

Table 55: PIV Auto APDU Reference: OPACITY / VCI Context

Operation	Command Shape	Key Ref	Relationship to PIV Auto
OPACITY establish	0087[27/2E]04[Lc]7C...81.. ..8200	04	Establishes VCI secure messaging and returns cryptogram/CVC evidence.
PIV Auto challenge	0087[alg][9A/9E][Lc]7C... 820081.....00	9A/9E	Performs the autonomous key challenge after card preparation and any required VCI setup.
Response handling	7C...82...9000	9A/9E	The PD reports the returned signed response bytes exactly, excluding only SW1/SW2.

D Deprecated osdp_GENAUTH

The legacy osdp_GENAUTH command is deprecated and expected to be removed in a future revision of this specification. Using a PIV credential to decrypt or process arbitrary data is unsafe and unnecessary for credential verification; implementations **SHOULD** use osdp_CRAUTH flows instead. Controllers and PDs **SHOULD** plan migrations away from osdp_GENAUTH.